EXPVID: A BENCHMARK FOR EXPERIMENT VIDEO UNDERSTANDING & REASONING

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ABSTRACT

Multimodal large language models (MLLMs) have achieved impressive performance on a broad range of video and image benchmarks, yet their ability to understand and reason about the fine-grained, long-horizon nature of real scientific experiments remains largely unexplored. We introduce ExpVid, the first benchmark dedicated to *experiment video understanding and reasoning*, built on 390 peer-reviewed laboratory videos sourced from the Journal of Visualized Experiments (JoVE) [JoVE], spanning 13 scientific disciplines with an average duration of 489 seconds per video. ExpVid evaluates MLLMs across a three-level hierarchy that mirrors the scientific workflow: **Level-1** tests fine-grained visual perception through four multiple-choice recognition tasks (Tool, Material, Quantity, and Operation); **Level-2** probes procedural understanding via step ordering, sequence generation, completeness verification, and step prediction; and **Level-3** demands holistic scientific reasoning through open-ended experimental analysis and scientific discovery questions. In total, ExpVid comprises 7,800 QA pairs across 10 tasks. All annotations are designed to be vision-centric, preventing shortcutting via language priors. We benchmark 19 state-of-the-art open- and closed-source MLLMs, revealing significant headroom: the best model (GPT-5 [OpenAI, 2025]) achieves only 57.5% on Level-2 and 56.4% on Level-3, while even strong models lag far behind on step prediction and sequence generation. ExpVid offers a rigorous testbed for building MLLMs capable of perceiving, verifying, and reasoning about real wet-lab experiments.

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1 Introduction

Scientific experimentation is a cornerstone of modern research, demanding careful observation of fine-grained visual details, strict adherence to procedural protocols, and integration of experimental evidence into broader scientific conclusions. For a model to truly assist scientists, it must not only identify what a researcher is doing at any given moment, but also track the full experimental workflow, detect missing or incorrectly ordered steps, and ultimately synthesize experimental observations into meaningful findings. These capabilities—spanning perception, procedure, and reasoning—remain far beyond the reach of even the most capable multimodal large language models (MLLMs) today.

Existing video benchmarks primarily target generic action understanding [Li et al., 2024, Mangalam et al., 2023, Caba Heilbron et al., 2015] or general-purpose knowledge tasks [Fu et al., 2025, Zhou et al., 2024, Yu et al., 2025]. Scientific benchmarks, on the other hand, tend to focus on static images or text [Yue et al., 2024, Lu et al., 2023, Wang et al., 2023, Saikh et al., 2022, Zhou et al., 2025], and the few video-based scientific benchmarks that exist [Zhao et al., 2025, Hu et al., 2025] do not probe the procedural fidelity and long-horizon scientific reasoning characteristic of wet-lab experiments. Prior work on fine-grained laboratory video datasets such as FineBio [Yagi et al., 2025] and SciVid [Hasson et al., 2025] addresses activity recognition in controlled settings but does not target the MLLM evaluation regime with hierarchical QA tasks. As a result, there is no benchmark that jointly evaluates fine-grained visual perception, procedural understanding, and scientific reasoning on authentic, peer-reviewed laboratory footage.

We introduce ExpVid (**Exp**eriment **Vid**eo Benchmark), designed to fill this gap. ExpVid is constructed from 390 peer-reviewed videos sourced from the *Journal of Visualized Experiments* (JoVE) [JoVE], a unique resource in which researchers publish complete protocols as rigorously documented videos alongside companion papers. This pairing of video, ASR transcripts, and published papers enables a rich three-level evaluation hierarchy:

- **Level-1 (Perception):** Multiple-choice recognition of tools, materials, quantities, and operations within short ($\sim$8 s) action-level clips ($\approx$10k clips).
- **Level-2 (Procedure):** Procedural understanding tasks—step ordering, sequence generation, completeness verification, and step prediction—on medium-duration ($\sim$48 s) stage-level segments ($\approx$3.5k segments).
- **Level-3 (Reasoning):** Holistic scientific reasoning over full experiments ($\sim$8 min), requiring open-ended experimental analysis and scientific discovery answers that cannot be inferred from text alone.

Our annotation pipeline combines LLM-assisted generation (DeepSeek-R1 [Guo et al., 2025a], Qwen2.5-VL [Bai et al., 2025b], GPT-5 [OpenAI, 2025]) with expert validation by PhD-level researchers in biology, chemistry, and medicine, ensuring vision-centric annotations that resist text-only shortcutting. We benchmark 19 MLLMs spanning open-source models [Bai et al., 2025b, Zhu et al., Wang et al., 2025a, Bai et al., 2025a, Team et al., 2025, Hong et al., 2025] and closed-source models [OpenAI, 2025, DeepMind, 2025b,a, Anthropic, 2025, Guo et al., 2025b] under both standard and chain-of-thought ("think") inference modes.

Our key findings include: (1) all models remain far from human-level scientific reasoning, with GPT-5 achieving 56.4% on Level-3 versus the significant challenge posed by full-experiment reasoning; (2) models show strong step ordering but struggle with step prediction and sequence generation, revealing a fundamental gap in prospective procedural reasoning; (3) thinking modes inconsistently improve performance—they help on perception but can hurt procedural tasks due to reasoning drift away from visual content; and (4) visual input is essential, with frame removal sharply degrading all levels, confirming the vision-centric nature of the benchmark.

In summary, our contributions are:

- The **first benchmark** targeting authentic experiment videos with a three-level evaluation hierarchy from fine-grained visual perception to holistic scientific reasoning.
- A **vision-centric, semi-automatic annotation pipeline** combining state-of-the-art LLMs with PhD-expert validation across 13 scientific disciplines.
- A curated dataset of **390 peer-reviewed experiments** with **7,800 QA pairs** spanning 10 tasks at action, stage, and full-experiment temporal scales.
- A **comprehensive evaluation of 19 MLLMs**, exposing critical gaps in procedural prediction and long-horizon scientific reasoning.

2 Related Work

General Video Benchmarks. A substantial body of work has established benchmarks for general video understanding. Short-video benchmarks like MVBench [Li et al., 2024] and EgoSchema [Mangalam et al., 2023] test action and scene comprehension, while ActivityNet [Caba Heilbron et al., 2015] and Charades-STA [Gao et al., 2017] focus on temporal grounding. Long-video benchmarks such as Video-MME [Fu et al., 2025] (averaging 1017 s), MLVU [Zhou et al., 2024], VRBench [Yu et al., 2025], and LVBench [Wang et al., 2024c] evaluate reasoning over extended temporal contexts. While these benchmarks cover a wide range of generic content, they do not address the procedural and scientific reasoning demands of laboratory experiments.

Scientific and Knowledge Benchmarks. MMMU [Yue et al., 2024] and MMMU-Pro evaluate expert-level multi-discipline understanding from static images; MMVU [Zhao et al., 2025] extends this to short videos. Video-MMMU [Hu et al., 2025] assesses knowledge acquisition across three cognitive stages using professional videos. MathVista [Lu et al., 2023], MathVision [Wang et al., 2024b], and SciBench [Wang et al., 2023] target mathematical and scientific problem-solving. ScienceQA [Saikh et al., 2022] covers broad K-12 science knowledge. SFE [Zhou et al., 2025] probes scientific signal perception and comparative reasoning across five disciplines. MMStar [Chen et al., 2024] focuses on vision-indispensable multimodal questions. These benchmarks, however, are primarily image-based or short-video-based, and do not evaluate the procedural understanding of multi-step laboratory workflows.

Procedural and Laboratory Video Understanding. FineBio [Yagi et al., 2025] introduced a fine-grained video dataset of biological experiments with hierarchical annotations (protocols, steps, atomic operations) for activity recognition. The work of Gabrieli et al. [2025] explores activity recognition in scientific experimentation using multimodal visual encoding. SciVid [Hasson et al., 2025] provides a cross-domain benchmark for spatiotemporal foundation models on scientific video tasks. While these efforts target recognition in controlled settings, they do not provide the comprehensive MLLM evaluation framework spanning perception, procedure, and scientific reasoning that ExpVid offers. ExpVid uniquely combines peer-reviewed JoVE footage with paired papers, ASR transcripts, and expert-validated, vision-centric QA across all three cognitive levels.

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Table 1: Comparison between ExpVid and some MLLM benchmarks. A and M indicate automatic and manual annotation, respectively.

Benchmark	#QA Pairs	#Videos	Avg. Sec.	#Task Types	Annotation	Domain
General Video Benchmarks						
MVBench [Li et al., 2024]	4,000	3,641	16.0	20	A+M	General
Video-MME [Fu et al., 2025]	2,700	900	1,017.9	1	M	General
MLVU [Zhou et al., 2024]	3,102	1,730	930.0	9	M	Narrative
VRBench [Yu et al., 2025]	9,468	960	5,796.0	1	M	Narrative
Knowledge-driven Benchmarks						
MMVU [Zhao et al., 2025]	3,000	1,529	51.4	2	M	Multi-disc.
Video-MMMU [Hu et al., 2025]	900	300	506.2	3	M	Multi-disc.
MathVision [Wang et al., 2024b]	3,040	–	–	1	M	Math
MathVista [Lu et al., 2023]	6,141	–	–	31	M	Math
MMMU [Yue et al., 2024]	11,500	–	–	2	M	Multi-disc.
ScienceQA [Saikh et al., 2022]	21,208	–	–	1	M	Science
SciBench [Wang et al., 2023]	789	–	–	1	M	Science
MMStar [Chen et al., 2024]	1,500	–	–	6	M	Multi-disc.
SFE [Zhou et al., 2025]	830	–	–	66	M	Science
ExpVid	7,800	390	489.0	10	A+M	Science

Multimodal Large Language Models. Recent MLLMs have demonstrated remarkable capabilities in visual and video understanding. Open-source models such as InternVL3 [Zhu et al.], InternVL3.5 [Wang et al., 2025a], Qwen2.5-VL [Bai et al., 2025b], Kimi-VL [Team et al., 2025], GLM-4.1V [Hong et al., 2025], and InternVideo2 [Wang et al., 2025b] have advanced video comprehension, while closed-source models including GPT-5 [OpenAI, 2025], Gemini-2.5-Pro [DeepMind, 2025b], Gemini-2.5-Flash [DeepMind, 2025a], and Claude-Sonnet-4 [Anthropic, 2025] lead performance on many general benchmarks. ExpVid provides a systematic evaluation of these models under scientific experiment conditions.

3 ExpVid: A Scientific Experiment Video Benchmark

3.1 Overview and Design Philosophy

ExpVid is designed to mirror the scientific workflow as a three-level hierarchy. Rather than evaluating isolated facts or static images, ExpVid requires models to engage with full experimental procedures at progressively larger temporal scales: from individual actions ($\sim$8 s) to procedural stages ($\sim$48 s) to complete experiments ($\sim$8 min). A key design principle is *vision-centricity*: all QA pairs are constructed to require visual observation and cannot be reliably answered from text priors or linguistic cues alone.

Figure 1 illustrates the three-level task hierarchy. Figure 2 provides an overview of the construction pipeline. A comparison to existing benchmarks is shown in Table 1.

3.2 Video Source and Selection

We source videos from the *Journal of Visualized Experiments* (JoVE) [JoVE], a peer-reviewed publication of scientific laboratory procedures in video format. JoVE videos are uniquely suited for our purposes: each video is accompanied by ASR transcripts and a rigorously peer-reviewed paper describing the experimental protocol, materials, and scientific findings.

Filtering. Raw JoVE videos are filtered using DeepSeek-R1 [Guo et al., 2025a] along five dimensions: Continuity (narrative flow), Alignment (video-transcript correspondence), Clarity (visual quality), Integrity (protocol completeness), and Focus (scientific relevance). Videos scoring below 4.0 overall or below 3.5 on any single criterion are discarded. Duration is restricted to the 25th–75th percentile range (378–728 s) to focus on complete, well-paced procedures.

Curation. Expert curators select 30 videos per discipline, yielding a final dataset of 390 videos across 13 scientific disciplines (including biochemistry, cell biology, molecular biology, neuroscience, immunology, genetics, developmental biology, microbiology, clinical medicine, materials science, chemistry, and environmental science).

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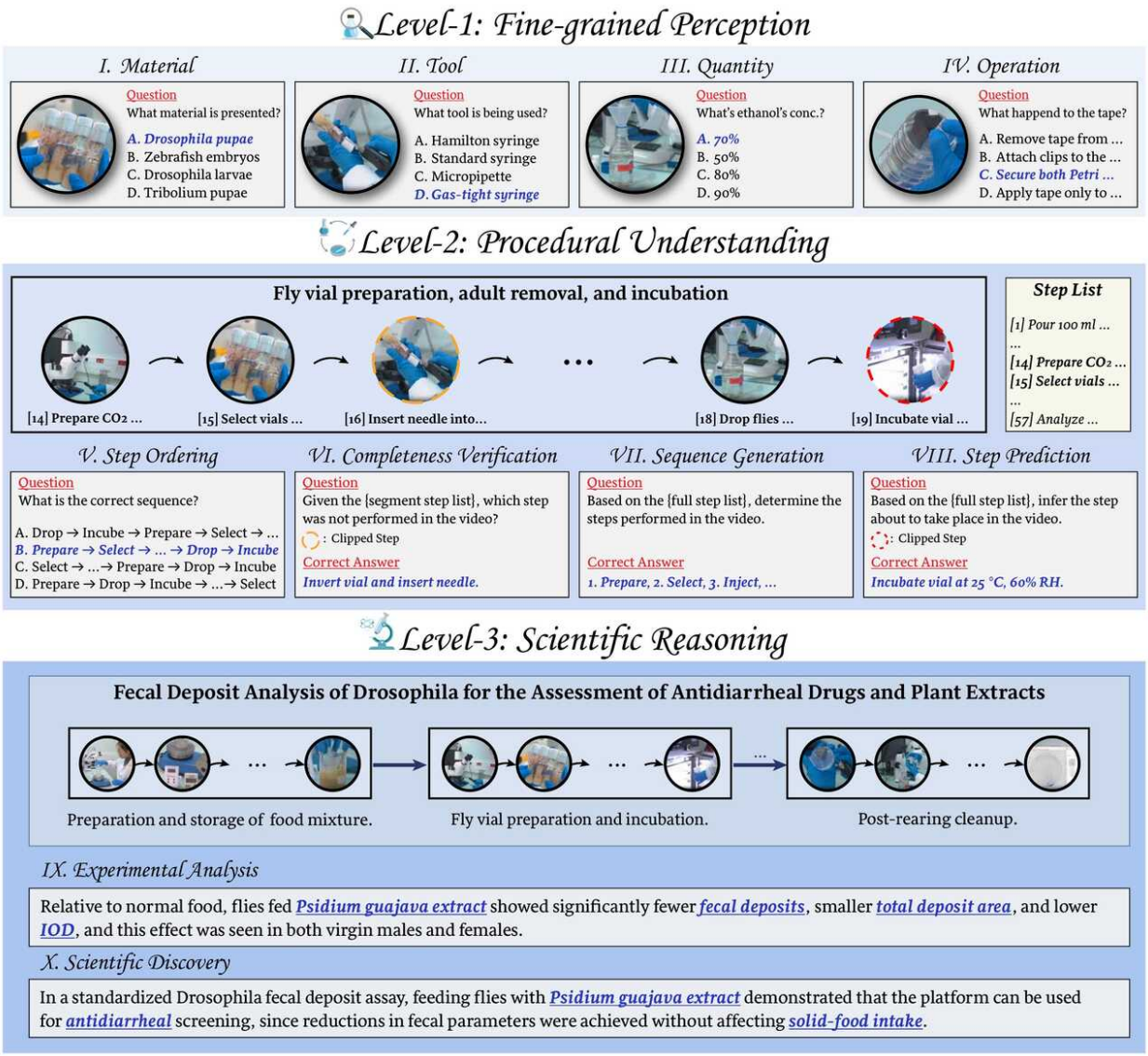

Figure 1: Illustration of the three-level task hierarchy in ExpVid. Level-1 covers action-level clips (∼8 s) for fine-grained perception; Level-2 covers stage-level segments (∼48 s) for procedural understanding; Level-3 uses full experiments (∼8 min) for scientific reasoning.

3.3 Task Definitions

ExpVid comprises 10 tasks across three levels, as illustrated in Figure 1.

Level-1: Perception (MCQ). Level-1 evaluates fine-grained visual recognition from short (∼8 s) action-level clips:

- **Tool Recognition:** Identify the laboratory instrument being used.
- **Material Recognition:** Identify the reagent, sample, or consumable.
- **Quantity Recognition:** Identify numeric quantities (volumes, concentrations, temperatures) shown or manipulated.
- **Operation Recognition:** Identify the specific laboratory action being performed (e.g., pipetting, centrifugation, staining).

All Level-1 tasks are formatted as 4-option multiple-choice questions (MCQ) evaluated by Top-1 accuracy.

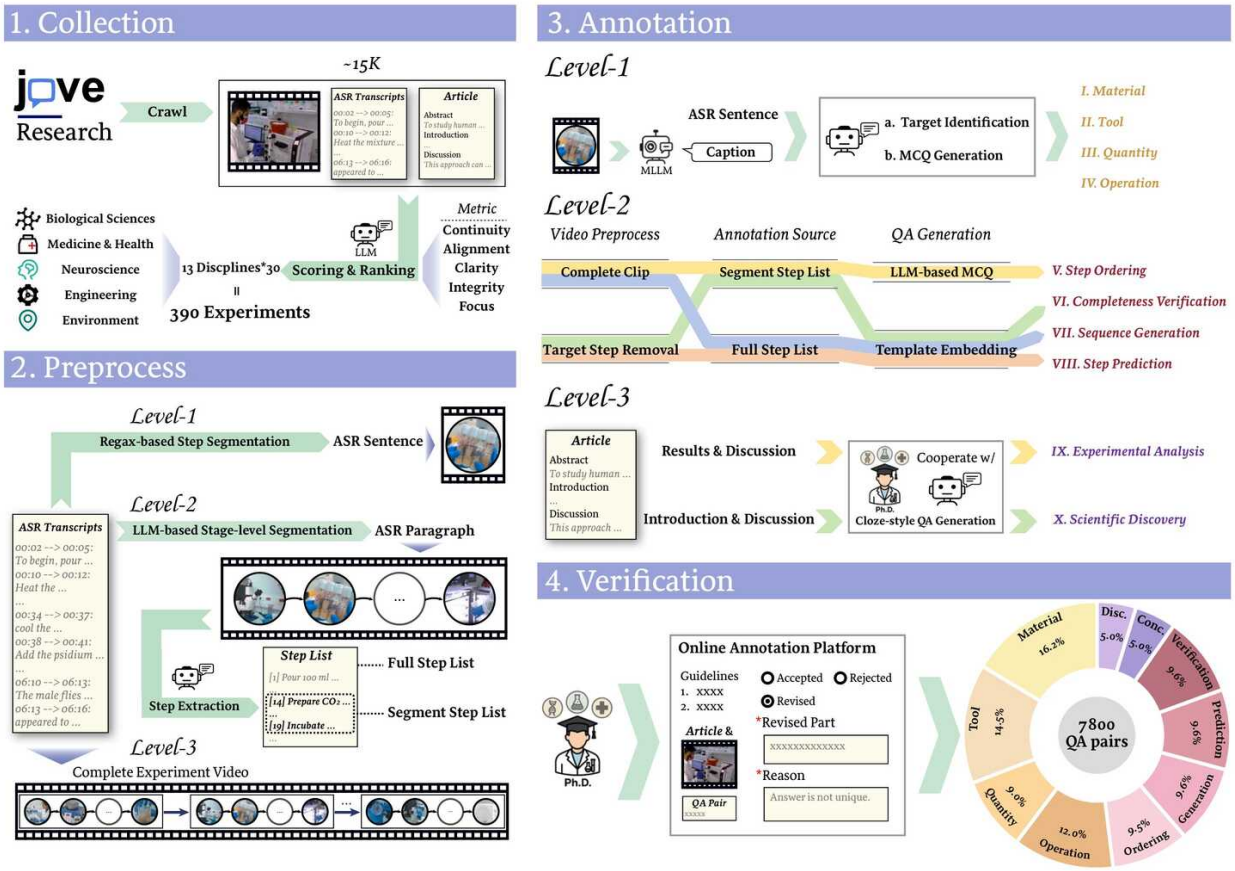

Figure 2: Overview of the ExpVid construction pipeline. Starting from JoVE videos, ASR transcripts, and paired papers, three parallel preprocessing tracks produce action clips (Level-1), stage segments (Level-2), and full-experiment inputs (Level-3). LLM-assisted annotation is followed by PhD-expert verification at each level.

Level-2: Procedure. Level-2 evaluates procedural understanding from stage-level segments (∼48 s):

- **Step Ordering:** Given a shuffled list of extracted steps, restore the correct temporal order. Evaluated by Jaccard similarity.
- **Sequence Generation:** Given a partial list of candidate steps, select all correct steps and arrange them in order. Evaluated by Jaccard similarity.
- **Completeness Verification:** Determine whether the shown sequence of steps is complete or has missing steps. Evaluated by Top-1 accuracy.
- **Step Prediction:** Given the current state of the experiment, predict the immediately following step. Evaluated by Top-1 accuracy.

Level-3: Scientific Reasoning (Cloze). Level-3 requires holistic reasoning over full experiments (∼8 min). Questions are formulated as cloze tasks with blanks requiring specific scientific knowledge:

- **Experimental Analysis:** Analyze experimental outcomes and explain mechanisms underlying observed results.
- **Scientific Discovery:** Connect experimental evidence to broader scientific conclusions or discoveries.

Level-3 answers are evaluated on a per-blank basis using Phi-3-mini [Abdin et al., 2024] as an LLM judge, measuring semantic equivalence to the ground-truth answer.

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3.4 Data Construction Pipeline

Preprocessing. For Level-1, videos are segmented by ASR sentence boundaries, producing approximately 10,000 action-level clips of $\sim$8 s each. For Level-2, stage boundaries are detected by DeepSeek-R1 [Guo et al., 2025a], requiring stage durations between 20–60 s, yielding approximately 3,500 stage-level segments of $\sim$48 s each. Step lists are extracted from the ASR and paper. For Level-3, the full experimental procedure video is retained; concluding slides and post-experiment data analysis sections are trimmed to prevent models from answering based on summary content rather than direct visual observation.

Level-1 Annotation. Target entities (tools, materials, quantities, operations) are extracted by DeepSeek-R1 [Guo et al., 2025a] from ASR transcripts. Visual visibility of each target in the corresponding clip is verified by Qwen2.5-VL [Bai et al., 2025b] through caption-based grounding. Four-option MCQs are then generated with carefully designed distractors: visually or functionally similar alternatives for tools and materials (e.g., a micropipette vs. a serological pipette), numerically proximate distractors for quantities, and visually confusable actions for operation recognition.

Level-2 Annotation. Step lists extracted from ASR and the companion paper are used to build procedural tasks. Step Ordering generates plausible permutations of extracted steps. Sequence Generation samples a candidate pool containing both correct and distractor steps, and models must select and order the correct subset. Completeness Verification inserts or removes steps to create positive and negative examples. Step Prediction asks models to identify the next step given the current procedural state.

Level-3 Annotation. Scientific paper sections are extracted using MinerU [Wang et al., 2024a]. Key anchor concepts are summarized by GPT-5 [OpenAI, 2025]. PhD-level domain experts (biology, chemistry, medicine) author cloze questions that require full-video reasoning and cannot be answered from text alone. An explicit vision-only answerability check ensures that each Level-3 question is genuinely unsolvable without observing the experiment video.

Expert Verification and Quality Control. All annotations undergo PhD-level expert review for visual visibility, distractor plausibility, step–segment alignment, and answer uniqueness. An iterative refinement process resolves ambiguities. The average annotation time per question is approximately 0.3 hours for Level-1, 0.5 hours for Level-2, and 1.2 hours for Level-3.

3.5 Dataset Statistics

ExpVid contains:

- **390 videos** across 13 scientific disciplines, average duration 489 s.
- $\approx$**10,000 action-level clips** ($\sim$8 s each) for Level-1 tasks.
- $\approx$**3,500 stage-level segments** ($\sim$48 s each) for Level-2 tasks.
- **7,800 QA pairs** in total across all 10 tasks.

Table 1 compares ExpVid to existing benchmarks. ExpVid is unique in combining scientific domain focus, authentic peer-reviewed video content, multi-scale temporal evaluation, and both automatic and manual (expert) annotation.

4 Experiments

4.1 Evaluated Models

We benchmark 19 MLLMs spanning a wide range of architectures and scales:

Open-source models: Qwen2.5-VL-7B and -72B [Bai et al., 2025b], MiMo-VL-7B-RL (w/ and w/o thinking), InternVL3-8B and InternVL3-78B [Zhu et al.], InternVL3.5-8B and InternVL3.5-38B [Wang et al., 2025a], Intern-S1 and Intern-S1-mini [Bai et al., 2025a], Keye-VL-8B-Preview and Keye-VL-1.5-8B, GLM-4.1V-9B and GLM-4.5V [Hong et al., 2025], Kimi-VL-A3B-Thinking [Team et al., 2025].

Closed-source models: Seed-VL-1.5 [Guo et al., 2025b], Claude-Sonnet-4 [Anthropic, 2025], Gemini-2.5-Flash (w/ and w/o thinking) [DeepMind, 2025a], Gemini-2.5-Pro (w/ and w/o thinking) [DeepMind, 2025b], GPT-5 [OpenAI, 2025].

Each model is evaluated with standardized prompt templates. Models supporting chain-of-thought reasoning are also tested in "think" mode (indicated by ✓ in Table 2).

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Table 2: Performance of evaluated models on the ExpVid across 10 tasks under three levels.

Model	Think	Level-1					Level-2					Level-3		
		Tool	Mat.	Quan.	Oper.	Avg.	Ord.	Gen.	Veri.	Pred.	Avg.	Anal.	Disc.	Avg.
Human Performance		17.5	15.9	61.3	55.5	37.6	69.8	31.2	45.6	21.8	42.1	–	–	–
Open-source MLLMs														
Qwen2.5-VL-7B-Instruct	×	32.0	33.9	49.0	62.4	42.6	56.2	20.8	20.7	1.3	24.6	25.2	21.4	23.3
MiMo-VL-7B-RL	×	34.2	33.7	44.2	62.4	42.4	43.9	28.5	18.5	11.4	27.4	28.7	25.9	27.3
MiMo-VL-7B-RL	✓	36.1	29.1	53.6	67.8	44.3	64.8	32.3	24.9	15.6	34.3	29.3	27.3	28.3
InternVL3-8B	×	27.5	31.0	38.8	65.6	39.4	43.4	20.4	20.2	3.9	23.9	29.2	25.3	27.2
InternVL3.5-8B	×	27.3	30.8	45.5	64.8	40.3	82.3	25.8	23.7	4.8	34.0	22.6	18.4	20.5
Intern-S1-mini	✓	33.3	31.2	52.5	61.4	42.5	73.6	14.3	16.8	8.3	28.1	33.5	28.3	30.9
Keye-VL-8B-Preview	✓	16.6	22.4	38.9	60.8	32.6	25.4	12.4	19.1	1.7	14.6	9.5	6.7	8.1
Keye-VL-1.5-8B	✓	21.0	23.4	51.3	64.0	37.0	56.7	9.5	20.0	2.8	22.1	8.4	6.1	7.2
GLM-4.1V-9B	✓	30.8	29.8	47.5	59.6	40.1	64.1	18.2	25.0	7.4	28.6	28.1	26.5	27.3
GLM-4.5V	✓	35.5	33.6	61.5	62.3	45.6	71.9	34.9	27.2	12.9	36.6	33.3	32.5	32.9
Kimi-VL-A3B-Thinking	✓	34.6	32.6	40.7	59.5	40.8	32.3	18.2	23.3	6.2	20.0	24.6	21.8	23.2
InternVL3.5-38B	✓	35.9	34.0	46.7	65.3	44.0	65.8	36.7	23.0	19.0	36.0	33.1	30.8	31.9
InternVL3-78B	✓	35.1	34.3	**73.2**	75.8	50.9	**87.1**	45.5	19.8	15.5	41.9	40.3	35.3	37.7
Qwen2.5-VL-72B-Instruct	×	30.5	34.7	54.5	64.5	43.9	86.3	34.1	23.8	0.3	35.9	31.9	29.3	30.6
Intern-S1	✓	38.9	35.2	58.9	73.8	49.9	82.2	45.0	24.1	15.4	36.0	43.0	36.3	39.6
Closed-source MLLMs														
Seed-VL-1.5	✓	32.9	24.6	43.9	69.2	40.7	73.9	48.6	19.8	27.9	42.5	32.0	29.4	30.7
Claude-Sonnet-4	×	25.6	31.2	54.3	61.9	40.8	78.7	37.6	16.5	11.6	36.0	29.1	30.1	29.6
Gemini-2.5-Flash	×	52.7	50.1	65.2	72.6	58.6	86.0	50.5	24.1	40.2	50.1	47.2	41.1	44.1
Gemini-2.5-Flash	✓	52.7	**50.7**	71.9	73.3	**60.2**	85.1	54.3	22.3	38.0	49.8	44.8	41.3	43.0
Gemini-2.5-Pro	×	**53.1**	45.9	64.3	**80.8**	59.2	83.7	61.3	**26.8**	49.6	53.8	50.6	45.2	47.9
Gemini-2.5-Pro	✓	51.3	44.3	63.8	74.4	56.7	84.2	59.9	**26.8**	46.9	54.3	50.1	44.8	47.4
GPT-5	✓	51.6	37.8	59.5	71.9	53.3	85.1	**66.9**	**26.8**	**51.8**	**57.5**	**55.4**	**57.4**	**56.4**

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4.2 Evaluation Metrics

- **Level-1** (Tool, Material, Quantity, Operation): Top-1 accuracy (%).
- **Level-2 Step Ordering and Sequence Generation**: Jaccard similarity between predicted and ground-truth step sequences (%).
- **Level-2 Completeness Verification and Step Prediction**: Top-1 accuracy (%).
- **Level-3** (Experimental Analysis, Scientific Discovery): Per-blank accuracy (%) as judged by Phi-3-mini [Abdin et al., 2024], measuring semantic correctness of each cloze blank.

Level averages (Avg.) are computed as unweighted means across the constituent tasks.

4.3 Main Results

Table 2 reports the full results for all 19 MLLMs. We provide human performance baselines for Level-1 (37.6) and Level-2 (42.1), collected from non-expert annotators who have not seen the videos before.

Closed-source models lead across all levels. GPT-5 [OpenAI, 2025] achieves the best Level-2 average (57.5) and Level-3 average (56.4), demonstrating superior procedural understanding and scientific reasoning. Gemini-2.5-Flash (think mode) [DeepMind, 2025a] leads Level-1 with 60.2, reflecting strong fine-grained visual perception. Gemini-2.5-Pro achieves competitive results at Level-1 (59.2) and Level-2 (53.8). Among open-source models, InternVL3-78B [Zhu et al.] achieves the best open-source results of 50.9, 41.9, and 37.7 on Levels 1, 2, and 3 respectively, but still falls substantially behind the leading closed-source models.

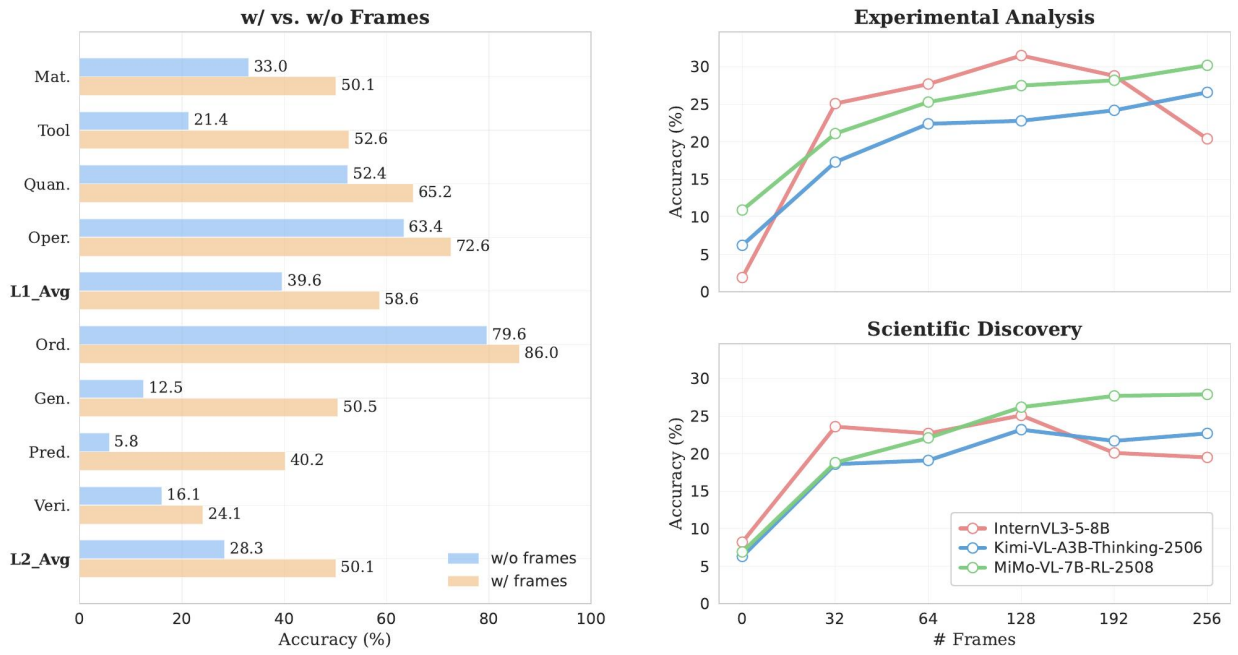

Figure 3: Effect of the number of input video frames on model performance across Level-1, Level-2, and Level-3. Removing frames (0-frame baseline) sharply degrades performance across all levels, confirming the vision-centric nature of ExpVid. Long-context video models (e.g., Kimi-VL) continue to improve with more frames (up to 256), while image-centric models saturate at lower frame counts.

Procedural asymmetry. Models exhibit a striking asymmetry within Level-2. Step Ordering, which requires only reordering a given set of steps, is relatively tractable: InternVL3-78B achieves 87.1 and Qwen2.5-VL-72B achieves 86.3. In contrast, Sequence Generation—which requires selecting and ordering steps from a mixed candidate pool—is far more challenging, with GPT-5 achieving only 66.9 and most other models scoring well below 50. Most strikingly, Step Prediction, which requires anticipating the next experimental action, proves extremely difficult for many models: Qwen2.5-VL-72B scores only 0.3, and most open-source models achieve single-digit scores. This asymmetry reveals a fundamental gap in prospective procedural reasoning: current models can retrospectively reorder known steps but struggle to predict what comes next from visual context alone.

"Thinking" is inconsistent. Chain-of-thought reasoning (think mode) does not uniformly help. For Gemini-2.5-Flash, think mode improves Level-1 performance (60.2 vs. 58.6) but slightly degrades Level-2 performance (49.8 vs. 50.1). For MiMo-VL-7B-RL, thinking improves Level-1 (44.3 vs. 42.4) and Level-2 (34.3 vs. 27.4). For Gemini-2.5-Pro, thinking slightly reduces Level-1 (56.7 vs. 59.2), while marginally improving Level-2 (54.3 vs. 53.8). This inconsistency suggests that extended chain-of-thought can cause reasoning drift away from visual evidence, particularly for procedural tasks requiring close attention to temporal visual cues.

Human performance. Human annotators (non-domain-expert) score 37.6 on Level-1 and 42.1 on Level-2. While closed-source models substantially surpass human non-expert performance on Level-1 (Gemini-2.5-Flash reaches 60.2), the gap narrows on Level-2, where GPT-5's 57.5 represents meaningful but not overwhelming advancement. Level-3 human baselines were not collected for this evaluation, reflecting the domain-expert knowledge required to assess free-form scientific conclusions.

4.4 Effect of Visual Input

To quantify the importance of visual input, we conduct ablation experiments varying the number of frames provided to models, ranging from 0 (text/transcript only) to the maximum supported by each model. Figure 3 shows the results.

Removing all visual frames sharply degrades performance on Level-1 and Level-2, and nearly nullifies Level-3 performance, confirming that ExpVid tasks cannot be solved from text priors alone. This validates our vision-centric annotation design: even Level-3 scientific reasoning questions require direct observation of the experimental video.

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With visual frames provided, performance generally increases with the number of frames, but the benefit depends on model architecture. Long-context video models such as Kimi-VL [Team et al., 2025] continue to improve with up to 256 frames, suggesting that temporal density matters for procedural understanding. Image-centric models, which encode video as collections of independent frames without dedicated temporal modeling, saturate earlier, typically around 32–64 frames, reflecting their limited capacity to integrate temporal information across a long experimental procedure.

4.5 Analysis and Discussion

Vision grounding at Level-1. Despite the short duration of Level-1 clips ($\sim$8 s), Tool Recognition and Material Recognition remain challenging for many models, with several open-source models scoring below 35% even when these categories require only identifying visible objects. Quantity Recognition is generally easier (73.2 for InternVL3-78B), suggesting that numeric visual cues are more tractable. Operation Recognition is typically the easiest subtask for capable models, as actions are visually salient in short clips.

Scientific reasoning at Level-3. Level-3 is the most challenging level. Even GPT-5 achieves only 56.4%, and most open-source models fall below 40%. The experimental analysis task (Anal.) and scientific discovery task (Disc.) are comparably difficult, with GPT-5 scoring 55.4 and 57.4 respectively. This indicates that models struggle both with mechanistic explanation (analysis) and with connecting observations to broader scientific significance (discovery). The gap between InternVL3-78B (37.7) and GPT-5 (56.4) at Level-3 is larger than at Level-1 (50.9 vs. 53.3), suggesting that long-horizon scientific reasoning is a domain where scaling closed-source systems yields larger gains.

Discipline diversity. ExpVid covers 13 scientific disciplines. While detailed per-discipline results are omitted from the main paper for brevity, our analysis finds consistent trends: models generally perform better on disciplines where visual cues are highly distinctive (e.g., materials science, chemistry) and worse on molecular biology and cell biology, where experimental steps can be visually similar despite being conceptually distinct.

5 Conclusion

We presented ExpVid, the first benchmark for evaluating multimodal large language models on authentic experiment video understanding and reasoning. Built on 390 peer-reviewed JoVE laboratory videos across 13 scientific disciplines, ExpVid offers 7,800 QA pairs across 10 tasks organized in a three-level hierarchy spanning fine-grained visual perception, procedural understanding, and holistic scientific reasoning. Our evaluation of 19 state-of-the-art MLLMs reveals substantial gaps across all three levels, with particularly large deficits in step prediction and scientific reasoning. Thinking modes provide inconsistent improvements, and visual input is essential across all levels. We believe ExpVid offers a rigorous and visually grounded roadmap for building MLLMs that can genuinely perceive, verify, and reason about real wet-lab experiments—a critical capability for AI-assisted scientific research.

Limitations and Future Work. ExpVid currently focuses on English-language JoVE videos, which may not cover the full diversity of global scientific practice. The Level-3 LLM-judge evaluation, while carefully validated, may not capture all nuances of scientific correctness. Future work could extend ExpVid to additional languages, wet-lab domains, and interactive settings where models must ask clarifying questions or propose experimental follow-ups.

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